

D207: Exploratory Data Analysis

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Section A: Research Question

A1. Research Question

Is **high blood pressure** correlated with a patient being **overweight**?

A2. Organization & Stakeholder Benefits

Relevancy To Our Organization

Being overweight elevates the risk of developing metabolic syndrome, characterized by risk factors like **high blood pressure** (John Hopkins Medicine, n.d.). Building on this existing medical research, which suggests a correlation between obesity and increased prevalence of high blood pressure, my study aims to investigate whether this association holds true within our own patient population.

I hypothesize that overweight patients admitted to our hospital will exhibit a higher incidence of comorbidities commonly linked to obesity, such as high blood pressure and the goal of my research question is to provide targeted evidence to either support or refute this hypothesis.

Reason for the Study

This analysis is potentially valuable to various stakeholders in our organization, including healthcare providers, policy makers, and health educators. Understanding if high blood pressure is associated with being overweight can help in:

- **Tailoring Prevention Programs:** By identifying this risk factor, our organization's health educators can design targeted interventions to prevent obesity and other, more serious, health conditions associated with high blood pressure, such as increased risk of heart disease, heart attack, and stroke (CDC, 2021).
- **Improving Patient Care:** Our healthcare providers can offer personalized advice and treatment plans for patients at risk of becoming overweight or developing high blood pressure due to obesity.
- **Informing Policy:** Our policy makers can use this information to develop public health policies and resource allocation strategies that address the root causes of obesity.

A3. Relevant Data

Defining The Research Question With Our Data

- **Overweight:**

Our data captures obesity (**overweight**) status as a categorical boolean variable rather than more specific individual body weights. While slightly limited, this factor is important for our hospital to consider due to the global public health burden of obesity, particularly in Westernized nations like the United States.

- **Health Conditions:**

Our data goes beyond just body weight to capture a broader spectrum of patient health. We track the presence of conditions like **stroke**, **diabetes**, **back pain**, **asthma**, **hyperlipidemia** (high cholesterol) and **high blood pressure**, recognizing the interconnectedness of these health conditions in shaping overall well-being. Focusing on these comorbidities is particularly important because:

- According to the Centers for Disease Control and Prevention (2022), “obesity-related conditions include heart disease, **stroke**, type 2 **diabetes** and certain types of cancer”.
- Obesity is a known contributing factor to both the development and aggravation of **back pain**. The additional weight can exert undue pressure on the spinal structures, leading to musculoskeletal imbalances and pain (Whelan, 2023).
- In regards to lung health, “the prevalence of **asthma** is increased in obesity, particularly in adult women” (Dixon & Nyenhuis, 2023).
-

Variables Within Our Dataset & **Variables Relevant To Our Research Question**

- Dependent Variable:
 - **‘Overweight’**: This column directly indicates whether a patient is overweight or not, making it the primary variable of interest for our research question. It is also a categorical variable.
- Independent **CATEGORICAL** Variables:
 - **‘Stroke’**: Indicates patient has a history of stroke
 - **‘Diabetes’**: Indicates patient has/does not have diabetes
 - **‘BackPain’**: Indicates patient has/does not have back pain
 - **‘Asthma’**: Indicates patient has/does not have asthma
 - **‘Hyperlipidemia’**: High cholesterol status
 - **‘HighBlood’**: High blood pressure status
 - **‘Gender’**: Patient gender identity as Male, Female, or Nonbinary
 - **‘Marital’**: Policyholder marital status
 - **‘Complication_risk’**: Patient’s risk of complications from Low to High
 - **‘Arthritis’, ‘Anxiety’, ‘Allergic_rhinitis’, ‘Reflux_esophagitis’**: Various other patient health conditions that my research hasn’t confirmed could be related to obesity
 - **‘Area’**: Urban, Suburban, Rural area type
 - **‘ReAdmis’**: Readmission status
- Independent **CONTINUOUS** Variables:
 - **‘Age’**: Patient’s age
 - **‘Children’**: Number of children in the patient’s household
 - **‘Income’**: Annual policyholder income
 - **‘VitD_levels’**: Patient’s vitamin D levels
 - **‘Doc_visits’**: Number of physician visits
 - **‘Full_meals_eaten’**: Number of full meals consumed by the patient during their hospitalization

- 'vitD_supp': Number of vitamin D supplements administered
- 'Initial_days': Initial number of days spent hospitalized
- 'TotalCharge', 'Additional_charges': Charges related to patient's healthcare

Data Preparation

- **Check for missing values:** To identify any missing values, the `isnull()` method is applied to all columns.
- **Feature selection:** Columns with high cardinality (many unique values) or those deemed irrelevant to the research question will be dropped using the `drop()` method. Categorical variables with low cardinality or those relevant to the research question will be retained.
- **Label standardization:** To improve data clarity and consistency, a new variable named `updated_labels` will be created. This variable will store an array of new column names that are standardized to lowercase, separated by underscores, and more accurately reflect the data they represent.

Section B: Data Analysis

B1. Python Code

See code attached via file 'd207_task1.ipynb'

B2. Data Analysis Output & Results

Statistical Analysis

This study employs a significance level of alpha (α) = 0.05 to determine statistically significant associations between overweight status and various health conditions. A p-value less than 0.05 indicates that the observed association is unlikely to be due to chance alone and thus, statistically significant.

Results

- **High Blood Pressure (HighBlood):** Our data reveals a statistically significant association between high blood pressure (HighBlood) and being overweight ($p = 0.00093$). This suggests a strong relationship between the two factors, meaning overweight patients are more likely to have high blood pressure, at least within this specific data set.
- **Other Health Conditions:** Contrary to the initial hypothesis, no statistically significant associations were found between being overweight and the other health conditions investigated, as their p-values were all greater than 0.05.

Out [4] :

	Chi2	p-value
HighBlood	6.763426	0.009304
Stroke	0.005411	0.941361
Diabetes	0.537066	0.463651
Hyperlipidemia	0.344400	0.557300
BackPain	0.972063	0.324166
Asthma	1.876961	0.170680

As shown in the table below, the prevalence of high blood pressure is significantly higher among overweight patients compared to those with normal weight. This observation aligns with the chi-square test results, which statistically confirmed a link between overweight status and high blood pressure.

Out [5] :

HighBlood	No	Yes	Total
Overweight			
No	1776	1130	2906
Yes	4134	2960	7094
Total	5910	4090	10000

B3. Data Analysis Technique Justification

Justification for Chi-Square Test Selection

For my research, a chi-square test was chosen over other methods for the following reasons:

- **Data Type & Suitability**
 - **Chi-square** tests are designed to analyze the **association between categorical variables**, examining if the observed distribution of the data aligns with expected distribution. In this study, the research question focused on the **association between being overweight (a categorical “yes/no” variable) and the presence or absence of various health conditions (also categorical**

“yes/no” variables). Therefore, the chi-square test was directly applicable to the data structure.

- **Comparison with Other Tests**

- **T-test:** Typically used to compare the means of two groups (independent or paired samples) for continuous outcome variables. If the objective was to compare the average values of a continuous variable (for example, average income) between overweight and non-overweight patients (assuming normality), a t-test would have been more appropriate.
- **ANOVA (Analysis of Variance):** Similar to t-tests, ANOVA assumes normally distributed, continuous data and compares means across 3 or more groups. Since all variables in this study are categorical, ANOVA was not suitable.

Objective Alignment

The research question aimed to identify an association between overweight status (categorical) and specific health conditions (categorical). As both the independent and dependent variables are categorical in nature, the chi-square test presented itself as the most appropriate option. It allowed for testing the hypothesis that the occurrence of these health conditions is independent of overweight status.

In summary, the chi-square test was chosen for its suitability in analyzing categorical data and its ability to test for associations between two categorical variables, aligning perfectly with the objective of my research question. T-tests and ANOVA are excellent for comparing means but simply do not fit the nature of my data and my research question being focused on categorical outcomes.

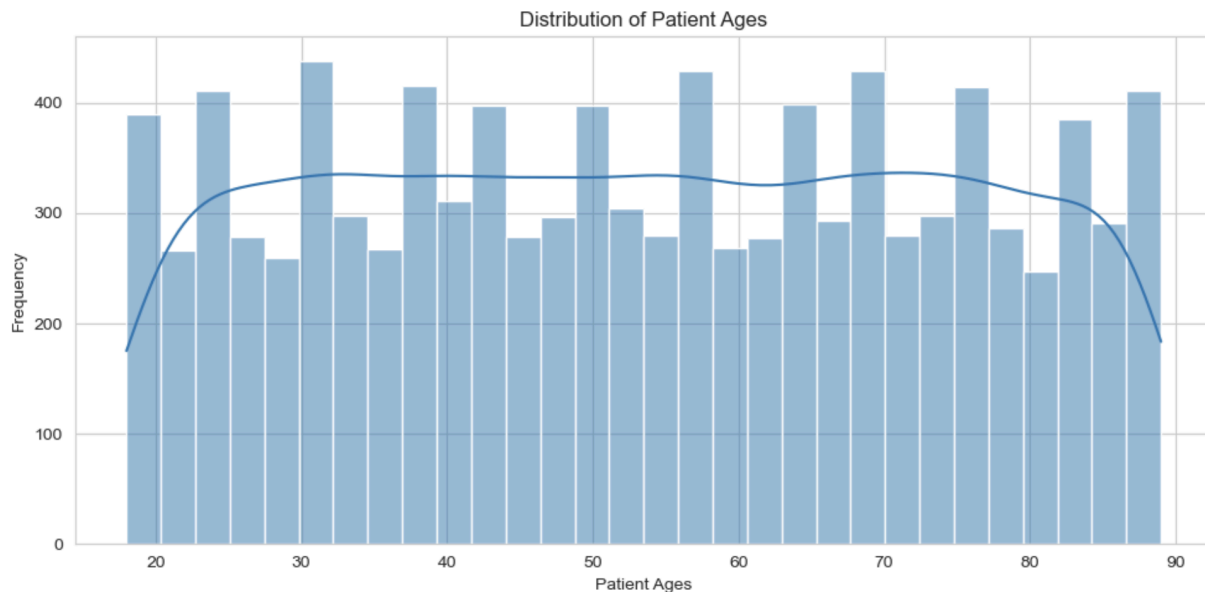
Section C: Univariate Statistics

See code attached via file 'd207_task1.ipynb'

C1. Visualizations (4)

See code attached via file 'd207_task1.ipynb'

Visualization #1: Distribution of Patient Ages

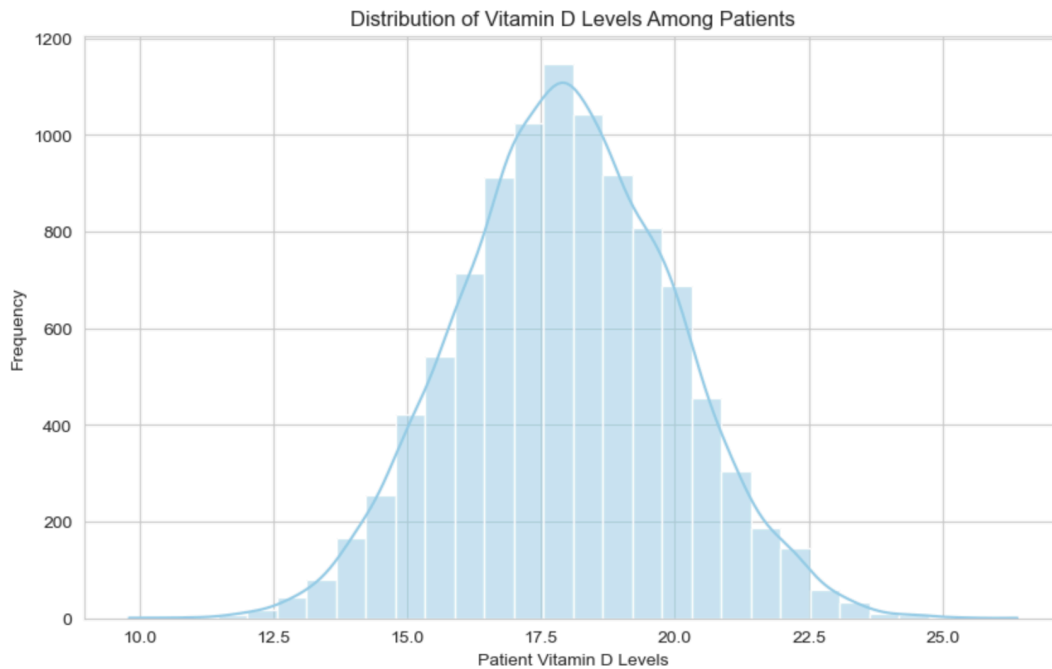


This histogram shows the distribution of ages among a group of hospital patients. The y-axis represents the number of patients (frequency) and the x-axis represents the patients' age. The smooth dark blue line represents the kernel density estimation (KDE) of the data, which helps visualize the overall distribution.

Key Observations

- **Distribution:** The histogram indicates that the patient ages aren't uniformly distributed. The data appear to be somewhat bimodal, with two peaks rather than just one, suggesting that there are two age groups that are more prevalent in this patient population.
- **Spread of Data:** The spread of the ages is quite broad, extending from 18 years old to 89 years old. This suggests a wide range of ages among the patients, indicating a diverse patient population.
- **Skewness:** There is a slight skew to the right, as the tail on the right-hand side of the histogram is longer than the left-hand side. This means that there are more older patients than would be expected in a perfectly symmetrical distribution.
- **Outliers:** There are no clear outliers visible in the histogram. Outliers would typically be seen as isolated bars at the ends of the distribution. However, the smooth decrease on the right tail suggests that there are progressively fewer patients as age increases, which is normal since, according to the Centers for Disease Control and Prevention (2022) the average life expectancy in the U.S. is about 77 years.

Visualization #2: Distribution of Vitamin D Levels Among Patients



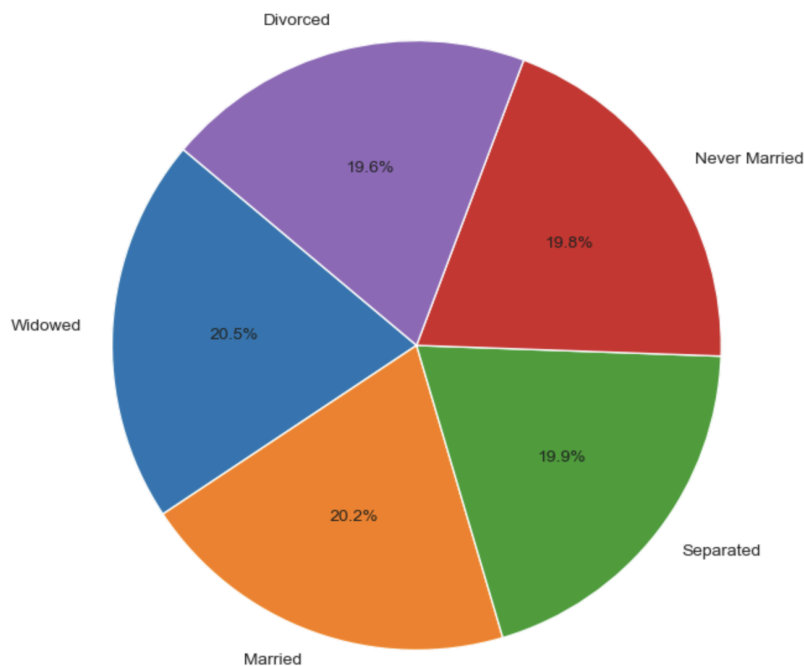
This graph shows the distribution of vitamin D levels among a group of hospital patients. The y-axis represents the frequency, and the x-axis represents the patients' vitamin D levels. The smooth blue line represents the kernel density estimation (KDE) of the data, which helps visualize the distribution's shape.

Key Observations

- **Distribution:** The distribution shows that vitamin D levels are approximately normally distributed, as indicated by the bell-shaped curve that is symmetric around the peak.
- **Spread of Data:** The spread of the data, shown by the width of the histogram, shows a moderate range of Vitamin D levels among the patients. The range is concentrated around the mean, which suggests that most of the patients have Vitamin D levels within a similar range.
- **Skewness:** There is no significant skewness visible in the histogram. Skewness would be indicated by a longer tail on one side of the distribution, but this histogram appears to be quite symmetric, suggesting that the mean and median of the Vitamin D levels are close to each other.
- **Outliers:** There are no visible outliers in the data, suggesting the vitamin D levels are relatively homogenous and that extreme vitamin D levels (either very high or very low) are uncommon. This could be useful for our hospital to understand that our patient population is within a certain range of vitamin D levels, as it can help in establishing reference ranges for normal levels or identifying atypical values for further investigation.

Visualization #3: Distribution of Marital Status Across Insurance Policyholders

Marital Status Distribution Across Insurance Policyholders

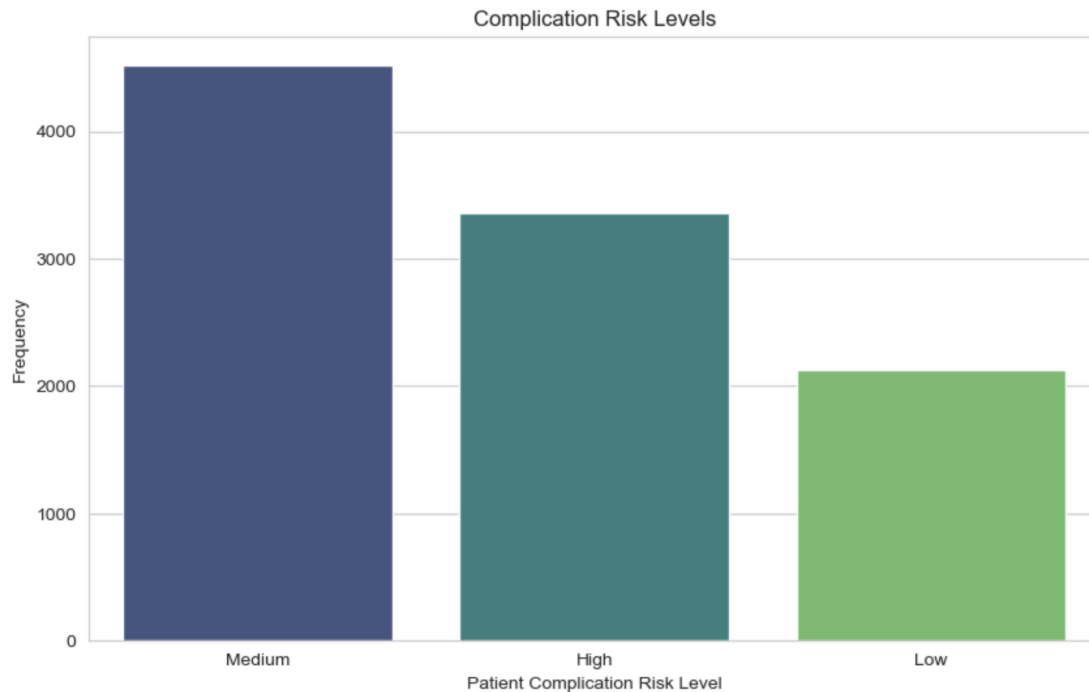


This graph shows the distribution of marital status among a group of insurance policyholders. The pie chart represents five categories: Married, Widowed, Divorced, Never Married, and Separated.

Key Observations

- **Distribution:** The distribution of marital status is relatively even, with each category accounting for roughly one-fifth of the total.
- **Spread of Data:** The spread of the data is represented by the size of each slice of the pie chart. Since the percentages are relatively close to each other, no single marital status category dominantly outweighs the others.
- **Interpretation:** The marital status of insurance policyholders is diverse and no single category is vastly more common than the others. The categories of 'Married' and 'Widowed' are the most common, each constituting over 20% of the policyholders, while the categories of 'Never Married', 'Divorced', and 'Separated' are also significant, each holding just under 20%. The data suggests that the insurance policyholders come from a variety of marital backgrounds.

Visualization #4: Distribution of Complication Risk Level



This bar chart depicts the frequency of patients with different levels of complication risk: High, Medium, and Low.

Key Observations

- **Distribution:** The bar chart displays the count of patients classified into three complication risk levels. The 'Medium' risk category has the highest frequency, followed by 'High', with 'Low' risk being the least frequent.
- **Spread of Data:** The data is distributed unevenly among the three categories, with more patients in the 'Medium' risk category than in 'High' or 'Low'.
- **Interpretation:** The most common complication risk level among patients is 'Medium', which could indicate that most patients have a moderate risk of complications. The 'High' risk category, while not as common as 'Medium', still represents a significant portion of patients, suggesting that a considerable group of patients might require more attention or resources due to their higher risk of complications. The 'Low' risk category has the least number of patients, which might suggest that fewer patients are at a low risk of complications or that the criteria to be considered at low risk are stricter or less common among the patient population. This info could be useful in determining resource allocation, planning for patient care, and risk management - knowing the distribution of complication risk can help with prioritizing patient monitoring and interventions.

Section D: Bivariate Statistics

See code attached via file 'd207_task1.ipynb'

D1. Visualizations (2)

See code attached via file 'd207_task1.ipynb'

Visualization #1: Distribution of Policyholder Income by Gender Identity



This violin plot displays the distribution of policyholder income segmented by patient gender identity: Male, Female, and Nonbinary.

Key Observations

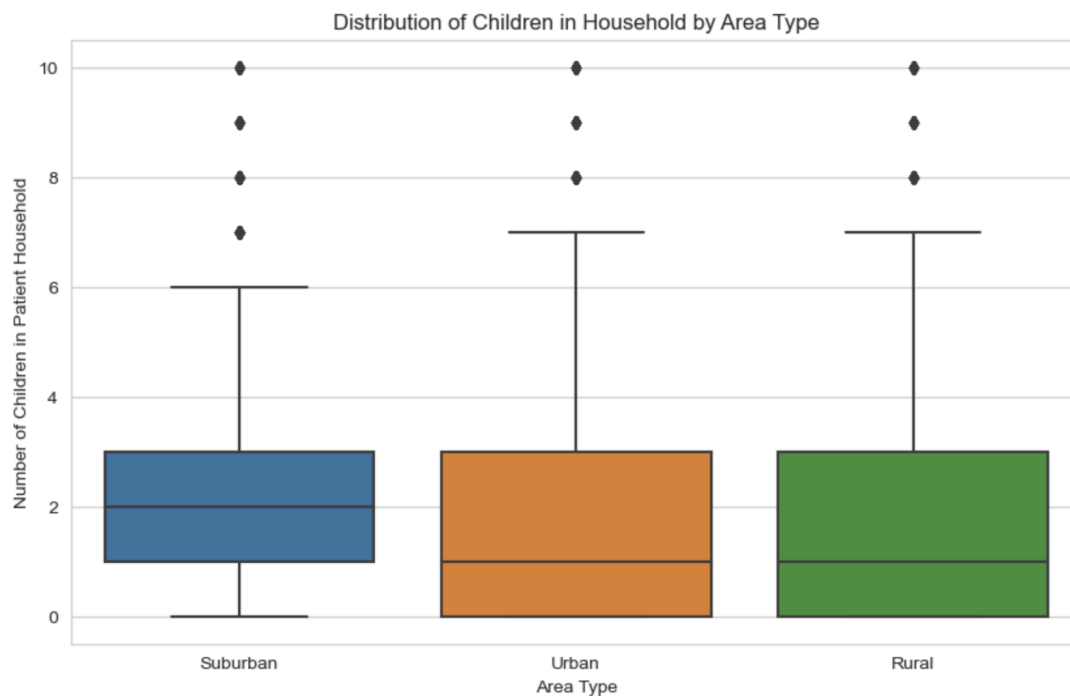
- **Median Income:** The center of each violin is marked by a little white dot, representing the median income for each gender group. The medians appear similar across all three categories, suggesting no significant differences in median income between genders.
- **Income Distribution:** The width of each violin indicates the distribution of policyholders across different income levels. Wider areas represent a higher concentration of policyholders at that income level. The violins for Male and Female show a wider base at lower incomes, tapering towards higher incomes. This suggests a higher proportion of policyholders in these categories have lower incomes. In contrast, the Nonbinary category has a more even distribution across income levels.
- **Spread of data:** The wider sections of each violin represent the interquartile range (IQR), which encompasses the middle 50% of the data. The IQR appears similar for Male and Female categories, but slightly narrower for the Nonbinary category, indicating a potentially smaller spread of incomes.

- **Skewness:** The tails of the violins for Male and Female genders extend further towards higher incomes, indicating a right skew. This means there are fewer policyholders in these categories with very high incomes, but a notable presence of them nonetheless. The Nonbinary category has a shorter tail, suggesting less skewness in their income distribution.
- **Outliers:** The lines that extend from the IQR towards the top and bottom of each violin indicate the range of the data excluding outliers. The points that lie beyond these lines are the outliers. There appear to be outliers in all categories, particularly at the upper end of the income range.
- **Comparison Across Genders:** While the median incomes are similar across genders, the distribution of income differs. The Male and Female distributions share a similar shape, with a wider base at lower incomes and a longer tail towards higher incomes. The Nonbinary distribution is more uniform, with fewer policyholders at both the very low and very high ends of the income spectrum compared to the other categories.

Interpretation

These findings suggest that although median incomes might be similar across genders, the distribution of income and the potential for higher incomes may be more pronounced among Male and Female policyholders. For policyholders identifying as Nonbinary, income levels appear to be more evenly distributed with fewer outliers on both ends of the spectrum.

Visualization #2: Distribution of Children in Household by Area Type



This box plot visualizes the distribution of the number of children in patient households across different area types: Suburban, Urban, and Rural.

Key Observations

- **Median Number of Children:** The line in the middle of each box represents the median number of children for each area type. Rural areas have a higher median compared to Suburban and Urban areas, indicating that families in rural areas tend to have more children on average.
- **Spread of Data:** The box itself represents the interquartile range (IQR), which encompasses the middle 50% of the data. The IQR for the Rural area is slightly shorter than the Suburban and Urban areas, suggesting less variability in the number of children among the typical rural households.
- **Range:** The "whiskers" of the box plot (the lines extending from the top and bottom of the box) show the range of the data within 1.5 times the IQR from the upper and lower quartiles. Suburban and Urban areas show a similar range, which is slightly less than the Rural area.
- **Outliers:** The dots above the whiskers represent outliers, which are data points that lie beyond 1.5 times the IQR from the quartiles. There are outliers in each area type, with Suburban having the least and Rural having the most. This indicates that there are households with significantly more children than typical for each area type, especially in Rural areas.
- **Skewness:** If we consider the median's position within the box and the length of the whiskers, it seems that all area types have a right-skew (positive skew), indicating that there are more households with a higher number of children than the median for each area.
- **Comparisons between Areas:** Rural areas not only have a higher median but also more outliers on the higher end, indicating that households with many children are more common in rural areas. Suburban and Urban areas have more similar medians and ranges, but Suburban areas have slightly fewer outliers, suggesting a slightly more uniform distribution of the number of children in households compared to Urban areas.

Interpretation

This visualization reveals a clear pattern - rural households tend to have more children compared to Suburban and Urban areas. This observation is supported by the higher median number of children and the presence of more outliers on the higher end in the rural area's distribution. Various factors, such as cultural norms, living conditions, and economic considerations, might influence this trend in family size across different areas.

Section E: Summary

E1. Results of Hypothesis

This study investigated the potential link between being overweight and various health conditions. We analyzed data on stroke, diabetes, high blood pressure, hyperlipidemia, back pain, and asthma.

Our analysis revealed a statistically significant association **only between being overweight and high blood pressure ($p = 0.0093$)**. This means that in our dataset, individuals who were overweight were more likely to have high blood pressure compared to those who were not overweight.

For the other health conditions considered (stroke, diabetes, hyperlipidemia, back pain, and asthma), the analysis did not identify statistically significant associations with being overweight ($p > 0.05$ for all).

Out[4]:

	Chi2	p-value
HighBlood	6.763426	0.009304
Stroke	0.005411	0.941361
Diabetes	0.537066	0.463651
Hyperlipidemia	0.344400	0.557300
BackPain	0.972063	0.324166
Asthma	1.876961	0.170680

To further explore the connection between overweight status and high blood pressure, a crosstab table was used to visualize the relationship that supported the statistical finding. The table showed a higher proportion of individuals with high blood pressure in the overweight group compared to the non-overweight group, further confirming the association between these two variables.

Out[5]:

HighBlood	No	Yes	Total
Overweight			
No	1776	1130	2906
Yes	4134	2960	7094
Total	5910	4090	10000

In conclusion, this study found a positive association between being overweight and high blood pressure within the analyzed data, but no significant associations with other health conditions considered.

E2. Limitations of Data Analysis

Sample Size

This analysis is based on a dataset containing 10,000 patient records. While this may seem like a large number, it represents a relatively small sample size within the context of a nationwide hospital network. A larger and more diverse sample encompassing various geographic regions and patient demographics would likely yield more robust results and allow for a more

comprehensive understanding of the relationship between overweight status and associated health conditions.

Oversimplification of 'Overweight' Category

The analysis relies on a binary classification of patients as either 'overweight' or not overweight. This approach lacks the necessary granularity for more nuanced analysis. The variable doesn't distinguish individuals who are mildly overweight versus those who might actually be morbidly obese. This lack of differentiation could lead to misleading interpretations and an underestimation of the potential health risks associated with varying degrees of excess weight. For instance, the health risks and implications of being just 2 pounds overweight are vastly different from someone carrying 200+ pounds of excess weight. Lumping them together undermines the severity of one and minimizes the potential dangers of the other.

Ambiguity in Health Condition Data

Our data lacks specific details regarding certain health conditions, hindering accurate analysis and interpretation of potential correlations. For example, the variable 'diabetes' doesn't distinguish between Type 1, Type 2, or gestational diabetes. These distinct types have varying etiologies and risk factors. The Centers for Disease Control and Prevention (2023) defines Type 2 diabetes as often linked to lifestyle choices and considers it potentially preventable through weight management, healthy diet, and exercise, whereas Type 1 diabetes is an autoimmune disorder unrelated to such lifestyle factors.

Similarly, 'asthma' without further categorization provides limited insight. While the risk of asthma increases with higher BMI, the specific type of asthma and its severity level are crucial factors in understanding the potential association with overweight status. Analyzing specific types of conditions instead of a single umbrella term would enable a more nuanced understanding of the relationship between overweight status and specific health complications.

E3. Recommendations

By collaborating with data analysts and healthcare professionals, we can refine our data collection methods and ensure that the information we collect is accurate, comprehensive, and standardized. This will empower us to leverage the full potential of data analytics and transform healthcare for the better.

By analyzing data trends, we can identify patterns and risk factors associated with various conditions. This knowledge can inform personalized care plans, preventative strategies, and targeted interventions. For example, our research identified a link between being overweight and high blood pressure. This finding can be used to develop educational programs for patients, highlighting the associated risks of high blood pressure, such as heart attack, stroke, and kidney complications (World Health Organization, 2023). This can help us help our patients make informed decisions and take proactive steps towards better health.

Recommendations

- Implement more refined classifications for overweight status:
 - Measure height and individual body weight
 - Calculate and record BMI by category
 - Category recommendations (Weir & Jan, 2023):
 - BMI <17: Severely Underweight
 - BMI 17-18.5: Underweight
 - BMI 18.5-24.9: Normal (healthy weight)
 - BMI 25.0-29.9: Overweight
 - BMI 30.0-34.9: Obese - Class I
 - BMI 35.0-39.9: Obese - Class II
 - BMI 40.0-49.9: Obese - Class III
 - BMI 50.0-59.9: Obese - Class IV
 - BMI 60.0-64.9: Obese - Class V
 - BMI >65.0: Obese - Class VI
- Collect more detailed information regarding specific types of health conditions for a more accurate and nuanced analysis of potential correlations:
 - 'Diabetes': Record the type
 - Category recommendations:
 - Type 1 Diabetes
 - Type 2 Diabetes
 - Gestational Diabetes
 - 'Asthma': Record the type and severity (American Lung Association, 2022)
 - Category recommendations for type
 - Allergic
 - Seasonal
 - Adult-Onset
 - Asthma-COPD
 - Exercise-Induced
 - Occupational
 - Category recommendations for severity
 - Intermittent
 - Mild
 - Moderate
 - Severe

Section F: Video Demonstration

Panopto video recording:

<https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=ca0484c3-f9f4-4560-b9fb-b12a000d1e56>

Section G: Data Source

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